ModelIt!

Student Activity Pack

The Energy Roller Coaster: Where Does Speed Come From?

How Energy Transforms, Transfers, and Gets Lost Along the Way

Grade 04L2 | NGSS 4-PS3-1, 4-PS3-4 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 If you let a marble roll down a ramp, it eventually stops. Where do you think the energy went?

2 What do you think friction is, and how does it affect moving objects?

3 Draw what you think happens to a roller coaster's energy as it goes from the top of a hill to the bottom and then along a flat track.

4 Do you think a roller coaster would go faster on a smooth track or a rough track? Why?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Hill Height | Track Friction | Potential Energy | Kinetic Energy | Heat Loss

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Smooth Track, Tall Hill

Hill Height at 80%, Track Friction at 10%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Rough Track, Tall Hill

Hill Height at 80%, Track Friction at 80%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Smooth Track, Short Hill

Hill Height at 20%, Track Friction at 10%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Loop Size — How big a loop in the track is — the coaster needs enough kinetic energy to make it through without falling

Air Resistance — The force of air pushing against the coaster as it moves — faster coasters experience more air drag

Coaster Mass — How heavy the roller coaster car and riders are — heavier cars have more energy at the same speed but also more friction

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If energy is never destroyed, why does a roller coaster eventually slow down and stop? Use evidence from your simulation to support your answers.

1 If energy is never destroyed, why does the roller coaster eventually stop?

2 How is rubbing your hands together related to what happens between the coaster and the track?

3 Why do roller coaster designers make the first hill the tallest one on the ride?

4 If you could build a track with zero friction, would the coaster ever stop? Why or why not?

5 How did your model help you understand the difference between what happens on a smooth track vs. a rough track?

8 STEM Challenge: Design the Friction-Fighting Coaster

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YOUR ENGINEERING MISSION

Design a marble track that minimizes energy loss to friction so the marble maintains the most speed from start to finish.

THE SCENARIO

A theme park is losing money because their roller coaster slows down too quickly and riders complain it is not exciting enough. Your engineering team has been hired to redesign the track to reduce friction and keep the coaster fast for longer. Use evidence from your model to make design decisions about hill height and track surface.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU TEST FOR THE SMOOTHEST TRACK SURFACE (FOIL, WAX PAPER, PLASTIC, CARDBOARD)?
2. HOW WILL YOU MEASURE HOW FAR THE MARBLE TRAVELS BEFORE FRICTION STOPS IT?
3. WHAT STARTING HEIGHT GIVES THE MARBLE ENOUGH ENERGY TO OVERCOME FRICTION ON EACH SURFACE?
4. HOW WILL YOU KEEP YOUR TEST FAIR BY CHANGING ONLY ONE VARIABLE AT A TIME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Mechanical Engineers design machines, vehicles, and rides by understanding how energy, friction, and motion work together. They design everything from roller coasters to electric cars. They earn $70,000–$115,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 A roller coaster starts at the top of a tall hill and rolls down to the bottom. What happens to its potential energy and kinetic energy during the ride down?

- **A.** Both potential energy and kinetic energy increase together
- **B.** Potential energy decreases while kinetic energy increases
- **C.** Potential energy increases while kinetic energy decreases
- **D.** Both potential energy and kinetic energy stay the same

2 In the model, Hill Height and Track Friction are external components. Why are they external?

- **A.** Because they are the biggest components in the model
- **B.** Because they are located outside the roller coaster car
- **C.** Because they are design choices we can control before the ride starts
- **D.** Because they do not affect the coaster's speed

3 The model shows that when Track Friction increases, Heat Loss increases, and Kinetic Energy decreases. What does this tell us about friction?

- **A.** Friction creates new energy that makes the coaster go faster
- **B.** Friction destroys the coaster's energy so it disappears completely
- **C.** Friction transforms kinetic energy into heat, leaving less energy for motion
- **D.** Friction has no effect on how fast the coaster moves

4 A student tests two roller coasters. Coaster A has a tall hill and a smooth track. Coaster B has the same tall hill but a rough track. Which coaster will be faster at the bottom, and why?

- **A.** Coaster B, because rough tracks push the coaster forward harder
- **B.** They will be the same speed because they start from the same height
- **C.** Coaster A, because less friction means less energy is lost to heat, so more energy stays as kinetic energy
- **D.** Coaster B, because friction adds extra energy to the system

5 The relationship between Heat Loss and Kinetic Energy is NEGATIVE. What does a negative relationship mean in this model?

- **A.** Heat Loss and Kinetic Energy both decrease at the same time
- **B.** When Heat Loss increases, Kinetic Energy decreases because energy going to heat cannot also be used for motion
- **C.** The relationship is bad for the roller coaster so scientists call it negative
- **D.** Heat Loss and Kinetic Energy are not connected to each other